

Exploring Structural Knowledge for Automated Visual Inspection of Moving Trains

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Abstract—Deep learning methods are becoming the *de-facto* standard for generic visual recognition in the literature. However, their adaptations to industrial scenarios, such as visual recognition for machines, product streamlines, etc., which consist of countless components, have not been investigated well yet. Compared with the generic object detection, there is some strong structural knowledge in these scenarios (e.g., fixed relative positions of components, component relationships, etc.). A case worth exploring could be automated visual inspection for trains, where there are various correlated components. However, the dominant object detection paradigm is limited by treating the visual features of each object region separately without considering common sense knowledge among objects. In this article, we propose a novel automated visual inspection framework for trains exploring structural knowledge for train component detection, which is called SKTCD. SKTCD is an end-to-end trainable framework, in which the visual features of train components and structural knowledge (including hierarchical scene contexts and spatial-aware component relationships) are jointly exploited for train component detection. We propose novel residual multiple gated recurrent units (Res-MGRUs) that can optimally fuse the visual features of train components and messages from the structural knowledge in a weighted-recurrent way. In order to verify the feasibility of SKTCD, a dataset that contains high-resolution images captured from moving trains has been

collected, in which 18 590 critical train components are manually annotated. Extensive experiments on this dataset and on the PASCAL VOC dataset have demonstrated that SKTCD outperforms the existing challenging baselines significantly. The dataset as well as the source code can be downloaded online (<https://github.com/smartprobe/SKCD>).

Index Terms—Automated visual inspection, deep convolutional neural networks (DCNNs), object detection, train component detection.

I. INTRODUCTION

RECENTLY, deep learning-based techniques have significant breakthroughs in object detection [1], [2]. Many state-of-the-art deep learning-based object detection methods use the region-based paradigm, which solves the classification and bounding box regression problem for each region independently with features extracted by deep convolutional neural networks (DCNNs) [1], [3]–[7]. However, some previous studies have showed that common sense and priori knowledge play an essential role in human object recognition as well [8], [9]. When identifying an object in a scene, reasoning by human common sense can help to make a correct recognition [10], [11]. Ignoring the common sense and priori knowledge may inevitably sacrifice the accuracy of object detection [12], [13]. Sabour *et al.* [14] introduced the capsule concept to explore object parts-whole relationships. A capsule is a group of neurons whose activity vector represents the instantiation parameters of a specific type of entity, such as an object or an object part.

Recent attempts of exploiting external knowledge for deep learning-based object detection can be mainly categorized into two lines. The first is to incorporate context around object or global scene context [12], [15], [16], while the second is to utilize object relationships and a feature-sharing mechanism [13], [17]–[20]. These two types of models capture different information. Moreover, there are different object relationships during different scene contexts, and they can be combined together to improve object detection performance. However, to the best of our knowledge, exploring these two cases together has not been well investigated yet.

Compared with the popular datasets, for example, ImageNet and COCO, components of industrial scenarios have some structural knowledge, for example, a wheel must contain a bearing on the normal operation of trains. Due to the limited datasets from industry, the study of exploring the structural knowledge for object detection is not widely investigated yet.

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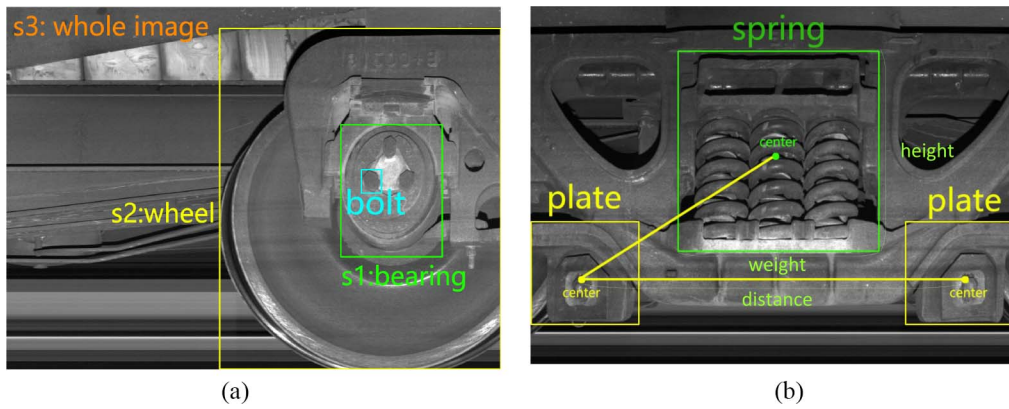


Fig. 1. Illustration of hierarchical scene context and object relationships in train images. (a) Hierarchical scene context. In this scene context, a bearing is likely to be detected, in which three bolts could be found further. For the bolts, the visual features of the entire image, the wheel, and the bearing can be regarded as hierarchical scene contexts. (b) Train component relationships with spatial dependencies. Each component may have relatively fixed positions and sizes.

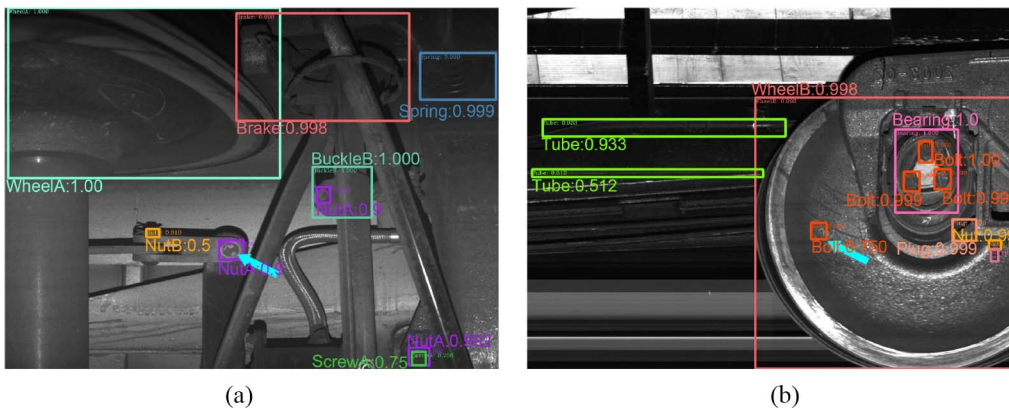


Fig. 2. Qualitative results on FPN [25]-based Faster R-CNN [1]. (a) Screw is not detected. (b) Bolt is mislabeled.

As for train-related scenarios, train component detection still poses some challenges in the industry especially in fault detection [21]–[24]. However, the public dataset for object detection and the work which explores structural knowledge for deep learning-based object detection in industrial scenarios are limited. To this end, we first release a dataset that contains 1130 images in a high resolution of 1400×1024 captured by high-speed line scan cameras from moving trains and a total of 18 590 critical components from 23 categories manually labeled by bounding boxes.

Structural knowledge of train components basically can be classified into two categories: 1) hierarchical scene context and 2) object relationships with spatial dependencies, which are shown as follows.

- 1) *Hierarchical Scene Context*: The semantic context among different components is in a hierarchical form. Take an example shown in Fig. 1(a) where a bearing is likely to be detected, in which three bolts could be further discovered. For the bolts, the visual features of the entire image, the wheel, and the bearing can be regarded as hierarchical scene contexts.
- 2) *Object Relationships With Spatial Dependencies*: Geometry dependencies among all train components may exist. As illustrated in Fig. 1(b), each component

may have relatively fixed positions and sizes. This may also be interpreted from a layout point of view. For example, the bearings are likely to be located at the center of train wheels. In this article, we show a nontrivial marriage of hierarchical scene context (e.g., global scene context and local scene context) and spatial-aware object relationships, via the proposed SKTCD, could lead to substantial improvement for train component detection. It is worth noting that such structural knowledge also widely exists in industrial scenarios, such as machines, product streamlines, etc., which consist of countless correlated components.

Fig. 2 shows two examples detected by the feature pyramid network (FPN) [25]-based faster region-convolution neural network (Faster RCNN) [1]. In Fig. 2(a), a nut is detected, while the small screw in this nut is not detected. If the local scene context of the screw (in this case is nut) can be taken into consideration, the accuracy of screws detection could be improved. Fig. 2(b) shows another example; a bolt is mislabeled because the faster R-CNN only concentrates on the object's visual appearance. With common sense, a bolt should not appear at that particular position of the wheel. The above detection error could be reduced

when the object relationships with spatial dependencies are considered.

In this article, we propose a novel framework, called SKTCD, which explores structural knowledge for automated visual inspection of trains. Our proposed model is portable and can be easily plugged into an other state-of-the-art two-stage-based object detection framework. We build SKTCD on the original faster R-CNN and FPNs for demonstration. Within SKTCD, the components are modeled into a hierarchical scene graph. Based on this constructed hierarchical scene graph, the detection of train components can be formulated as not only a visual recognition but also a graphical reasoning problem. Specifically, the visual features of nodes in the graph are integrated with messages from the hierarchical scene context and object relationships with spatial dependencies by our proposed residual multiple gated recurrent unit (Res-MGRU) in a weighted-recurrent way. Finally, the eventual integrated node features are used to predict object categories and bounding box offsets. By exploring the structural knowledge of train components, the proposed SKTCD outperforms the challenging baselines of object detection significantly. It is worth noting that hierarchical scene context and object relationships with spatial dependencies also exist in the dataset for general object detection. Experiments on the PASCAL VOC dataset also show the superior performance of our proposed SKTCD.

Our contributions to this article are summarized as follows.

- 1) We release a new dataset for train component detection that contains 1130 images and a total of 18590 components from 23 categories manually annotated with bounding boxes.
- 2) We design a novel Res-MGRU in SKTCD, which can fuse visual appearance, hierarchical scene context, and object relationships with spatial dependencies of the components jointly in a weighted-recurrent way, to boost the detection performance.
- 3) We propose an end-to-end framework, SKTCD, which considers train component detection as both visual recognition and graphical reasoning problems beyond DCNNs, by exploring the structural knowledge among train components.

II. RELATED WORK

A. Object Detection

Given an image as input, object detection aims at determining whether there are some instances of objects from a predefined category set and, if present, to return the bounding box of each instance [1], [26], [27]. With the renovation of deep learning, DCNNs reign undisputed as the new *de-facto* method for this task. Under this umbrella, existing deep learning approaches can be categorized into the two following families.

One-Stage Deep Learning-Based Detector: Since the pioneering work of OverFeat [28], many advanced one-stage deeply learned detectors have been introduced. SSD [29] modeled the output space of bounding boxes based on a set of default boxes over different aspect ratios and scales per feature map location. YOLO [30], [31] regarded object detection as a

regression problem to spatially separated bounding boxes and associated class probabilities. RetinaNet [32] addressed the issue of foreground-background class imbalance by reshaping the standard cross-entropy loss to down weight the loss assigned to well-classified examples.

Two-Stage Deep Learning-Based Detector: Although faster and simpler, one-stage detectors are reported to perform less accurate than their two-stage counterparts [1], [33]. SPP-net [34] utilized the concept of “spatial pyramid pooling” to handle images or subimages of an arbitrary size/scale. Fast RCNN [35] extended RCNN into a single-stage training with a multitask loss. In Faster RCNN [1], a region proposal network (RPN) that shares full-image convolutional features with the detection network was introduced to enable nearly cost-free region proposals. Other extensions of RCNN can be found in [25] and [33].

B. Knowledge-Based Recognition

Knowledge-based recognition in deep learning is to extract the background knowledge in order to improve recognition accuracy. Many of the previous works only focused on the image classification problem [36], [37]. Recently, some works were proposed to apply knowledge in the existing object detection frameworks. Fang *et al.* [38] adopted external knowledge to adjust the object detection scores obtained by the object detection framework. HKRM [20] incorporated the reasoning routed by two kinds of knowledge forms: an explicit knowledge module for structured constraints that is summarized with linguistic knowledge. In this article, the knowledge here mainly refers to the hierarchical scene context and object relationships with spatial dependencies as discussed before. Our main focus is to integrate knowledge into the proposed object detection framework in an end to end and implicit style. The approaches which explore scene context and object relationships for object detection are discussed in the following.

C. Exploring Scene Context for Object Detection

In real-world scenarios, visual objects usually coexist with contexts, which play an essential role in human object recognition [10], [11]. In DeepIDNet [16], results of image classification and object detection were combined together to improve the performance of object detection. In [12], spatial recurrent neural networks (RNNs) were utilized to explore contextual information across the entire images. Furthermore, some work also explored object relationships to improve object detection performance. A structure inference network (SIN) was introduced in [13], which integrated the global scene context with some relationships in an iterative reasoning way. However, this method ignores the local scene context, which is one of the important factors to improve object detection performance, especially in the industrial scenarios as discussed before.

D. Exploring Object Relationships for Object Detection

Object detection methods usually ignore object relationships, which exist in the real world. Recently, some work has utilized object relationships to improve the performance of

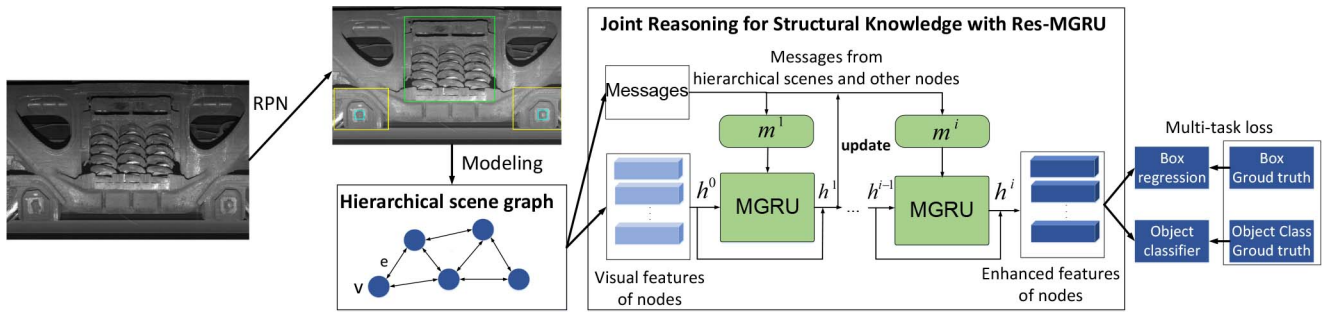


Fig. 3. Overview of our proposed SCTD, which is built on Faster RCNN for demonstration. First, the object proposals generated by the RPN are modeled into a hierarchical scene graph. Second, based on the constructed graph, the visual features of each object proposal are iteratively fused with the hierarchical scene context, visual features, and spatial information from other object proposals by our proposed Res-MGRU.

object detection. Graph R-CNN [39] was reported to be both effective and efficient at detecting objects and their relations in images with a novel relation proposal network (RePN). An iterative message-passing strategy was employed in [40] to explicitly model the objects and their relationships. An example of using a knowledge graph for image classification can be found in [36]. Relation network (RN) [18] was proposed to process a set of objects simultaneously through interaction between objects based on the attention mechanism.

III. OUR PROPOSED FRAMEWORK

In this section, we first give an outline of the original Faster RCNN. Then, we present the overview of our proposed framework, which is built on Faster RCNN and FPN for demonstration. Finally, how to exploit the structural knowledge to improve the performance of train component detection with our proposed Res-MGRU is presented.

A. Description of Faster RCNN

The inputs of Faster RCNN are the original images with any size while the outputs are bounding boxes with classification scores, which denote where are the detected objects and what are the objects. Faster RCNN [1], [41] consists of two stages. The first stage, called an RPN, proposes candidate object bounding boxes. An RPN takes an image (of any size) as input and outputs a set of rectangular object proposals, each with an object score [41]. RPNs are designed to efficiently predict region proposals with a wide range of scales and aspect ratios and can be trained in an end-to-end manner. The generated proposals are fed into the region-of-interest (RoI) pool, where each proposal consists of a feature vector and the positions of the proposal. The second stage, which is in essence Fast RCNN [35], extracts features using the RoI pool from each candidate box and performs classification and bounding-box regression. In the original Faster RCNN, region proposals, which are generated by the RPN, are classified with visual appearance clues only and are processed independently, which ignores the structural knowledge.

B. Framework Overview

In this article, we propose a framework, called SKTCD, on Faster RCNN to leverage the structural knowledge in order to

improve the object detection performance. Different from the original Faster RCNN, our framework is designed to take the additional hierarchical scene context and object relationships with spatial dependencies into consideration.

As shown in Fig. 3, we build SKTCD based on Faster RCNN [1] as a demonstration. Taking an image as input, region proposals that may contain some objects are generated through the RPN [1]. Then, the proposals are modeled into the hierarchical scene graph. Subsequently, the visual features of nodes, messages from hierarchical scene context, and spatial-aware object relationships are fused together recurrently in a gated way using our proposed Res-MGRU. Finally, the eventually integrated node features are used to predict object categories and bounding box offsets. The framework is end-to-end trainable, where the inputs of SKTCD are the original images while the outputs are the images with bounding boxes and object categories. Our proposed Res-MGRU is portable and can be injected into other state-of-the-art two-stage-based object detection frameworks.

To improve the detection performance of small train components, we utilize the FPNs [25] in the backbone networks of Faster RCNN. The construction of an FPN involves a bottom-up pathway, a top-down pathway, and lateral connections. FPN leverages the pyramidal feature hierarchy of convolutional networks (that has semantics from low to high levels) and builds a feature pyramid with high-level semantics throughout to improve the object detection performance especially for small objects. We discuss the proposed framework, SKTCD, in detail as follows.

C. Modeling

In this section, we discuss the modeling phase for hierarchical scene context and spatial-aware object relationships as depicted in Fig. 4. There are two subprocedures in the modeling process. First, the object relationships with spatial dependencies are modeled with graphs. Then, the hierarchical scene context is integrated to the graphs.

1) *Spatial-Aware Object Relationships With Graphs:* Applying RPN [1] to the input images, thousands of region proposals that may contain objects can be generated. Nonmaximum suppression (NMS [42]) is then utilized to select a fixed number of ROIs. Based on the region proposals, we can build a full region-to-region directed graph,

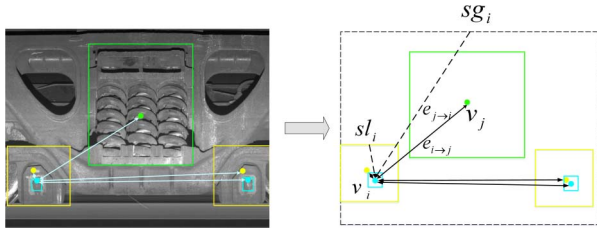


Fig. 4. Modeling the hierarchical scene context and spatial-aware object relationships with the hierarchical scene graph.

$G = (V, E)$, in which $v \in V$ is a node which denotes a region proposal and $e \in E$ is the edge (i.e., showing relationship) of each pair among object proposal nodes. The formulation of V and E is described in the following. For each object proposal node v_i , the spatial feature of each object proposal node is also stored. Therefore, each object proposal node can be denoted as $v_i = (sn_i)$, where sn_i denotes the spatial information of the node.

Given that a directed edge $e_{i→j}$ flowing from node v_i to v_j , we use both the spatial information and visual feature of v_i and v_j to compute a scalar, which represents the influence of v_i on v_j . Therefore, the relationships among object proposal nodes with visual features and spatial information can be modeled into a graph. Motivated by the observation that such an object-object relationship $e_{i→j}$ is also related to the relative component position and visual features, for example, a wheel generally contains a bearing in the middle and the bearing is more important to the wheel than other components nearby, we jointly consider the relative object positions and visual features. We can extract the visual features fn_i by a fully connected (FC) layer for node v_i . The object-object relationship $e_{i→j}$ as a scalar weight that represents the influence of v_i on v_j is modeled as follows:

$$e_{i→j} = \text{relu}(W_p R_{i→j}) * \tanh(W_v [fn_i, fn_j])$$

$$R_{i→j} = \left[w_i, h_i, s_i, w_j, h_j, s_j, \frac{x_j - x_i}{w_j}, \frac{y_j - y_i}{h_j} \right] \quad (1)$$

where W_p and W_v are the learnable weight for the relation of position and visual features, $R_{i→j}$ is the relative position of objects i and j , while w , h , and s are the width, height, and area of a proposal node, respectively, and (x, y) is the coordinate of the center of a proposal node.

2) *Integration of Hierarchical Scene Context*: Now, we discuss the way to integrate hierarchical scene context into the constructed graphs. For simplicity, we consider two tiers of scene context: 1) global scene context; and 2) local scene context. More tiers of scene context can be considered similarly. As shown in Fig. 4, node v_i is associated with both the global scene context and local scene context. Therefore, in each node v_i , we can form $v_i = (sn_i, sg_i, sl_i)$, where sn_i denotes the spatial information for node v_i , sg_i is the spatial information of the global scene context of v_i , and sl_i is the spatial information of the local scene context of v_i .

The way of finding the spatial information of the global scene context and the local scene context for each node is described as follows. For sg_i of node v_i , we utilize the entire image as the global scene context. For sl_i of node v_i , we first

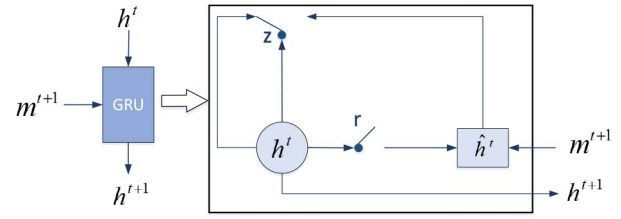


Fig. 5. Illustration of GRU at the iteration step t .

find the smallest proposal node v_k , which includes node v_i . If the proposal node cannot be found, a region which is a little bigger than v_i is utilized.

D. Recurrent Gated Fusion With Gated Recurrent Unit

RNNs [43] have recently shown promising results in many machine-learning tasks, especially when the input and/or output are a sequence of variables. Gated recurrent unit (GRU) is a simple and yet powerful variant of RNNs for time-series prediction due to the gated mechanism [44], [45]. GRUs are carefully designed to memorize historical information and fuse current states, new inputs, and historical information together in a recurrently gated way.

Based on the constructed hierarchical scene graph, we utilize GRU to recurrently fuse visual features of nodes with messages encoding from hierarchical scene context and other nodes together. In SKTCD, the original hidden states are set as the original visual features of all the nodes in the constructed graph. In each iteration of GRU, the current states are fused with incoming messages in a gated way. The fusion processes are conducted iteratively. After an iteration, the hidden states of nodes will be changed. When the hidden states of the nodes are updated, the hierarchical scene contexts and relationships among objects will also be updated. In the following iteration, GRU takes the updated hidden states as their new hidden states, and also takes the new messages from hierarchical scene context and other nodes as inputs, then computes the next hidden states. The updated mechanism of the messages is discussed in Section III-E. More updating can make the model more stable. Finally, the eventual integrated hidden states of all the nodes are used to predict object categories and bounding box offsets.

Formally, the original hidden states h_0 are defined as follows:

$$h^0 = \{fn_1, \dots, fn_n\} \quad (2)$$

where $fn_1, \dots, fn_n$ are the visual features of all the nodes, respectively, and n is the number of proposals. The computing process of the GRU unit at time $t + 1$ can be shown in Fig. 5 and formulated as follows:

$$r^{t+1} = \sigma \left(\Theta_r \left[m^{t+1}, h^t \right] + b_r \right)$$

$$u^{t+1} = \sigma \left(\Theta_u \left[m^{t+1}, h^t \right] + b_u \right)$$

$$c^{t+1} = \tanh \left(\Theta_c \left[m^{t+1}, \left(r^{t+1} \odot h^t \right) \right] + b_c \right)$$

$$h^{t+1} = u^{t+1} \odot h^t + \left(1 - u^{t+1} \right) \odot c^{t+1} \quad (3)$$

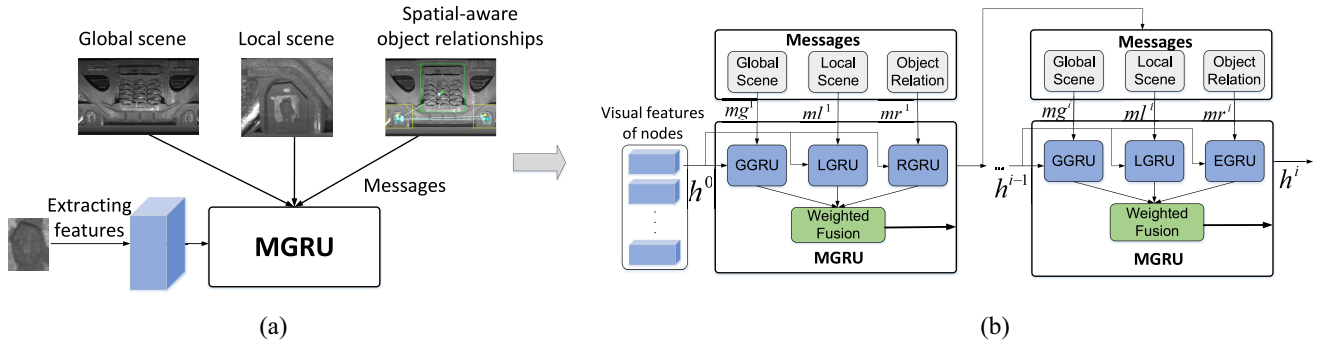


Fig. 6. Joint reasoning with MGRU. The visual features, messages from the global scene context, the local scene context, and the spatial-aware object relationships are exploited jointly by the MGRU. (a) Taking the joint reasoning of a specific node (a proposal for a nut) based on MGRU as an example. (b) Details of the MGRU. The inputs of the MGRU are the nodes in the hierarchical scene graph. An MGRU contains a GGRU, an LGRU, and an RGRU, which fuse the visual features of nodes with the messages from the global scene context, the local scene context, and the messages from the spatial-aware object relationships, respectively. A weighted fusion scheme is utilized in MGRU to integrate different kinds of messages.

where h^t is the hidden state of a GRU generated at the iteration step t and is original hidden states for the iteration step $t + 1$, m^{t+1} is the incoming message, h^{t+1} is the generated hidden state of a GRU, r^{t+1} and u^{t+1} are the reset gate and update gate at time $t + 1$, respectively, Θ_r , Θ_u , and Θ_c are the learned parameters of filters, and $\odot$ is the elementwise multiplication of tensors.

At iteration step $t + 1$, GRU can fuse h_t and m^{t+1} in a gated way and then generate the new hidden states h^{t+1} . We can let m^{t+1} be the information from the hierarchical scene context or from other nodes. Through some iterations, the original visual features of all the nodes and the information from the hierarchical scene context or from other nodes are fused iteratively. The memory cell can control how much information from the previous state will carry over to the current hidden state [44], thus allowing a more compact representation through the update gate u .

E. Joint Reasoning for Structural Knowledge With Multi-GRU

In SKTCD, we fuse the visual features of object proposals with messages from the hierarchical scene context and spatial-aware object relationships together. However, the disadvantage of GRU is that it can receive only one type of incoming message. To overcome this, we propose a novel MGRU to receive multiple types of incoming messages and fuse them together. Again, we take two tiers of scene context as an example (global scene context and local scene context), as shown in Fig. 6. MGRU contains three GRUs: 1) GRU for global scene context (GGRU); 2) GRU for local scene context (LGRU); and 3) GRU for object relationships (RGRU). The initial hidden states of MGRU are the visual features of all the nodes in the constructed graph. Each GRU unit takes one kind of messages as input to update current states of nodes iteratively. Taking the RGRU as an example, we can formulate the process of an RGRU unit as follows:

$$\begin{aligned} rr^{t+1} &= \sigma\left(\Theta_{rr}\left[mr^{t+1}, h^t\right] + b_{rr}\right) \\ ru^{t+1} &= \sigma\left(\Theta_{ru}\left[mr^{t+1}, h^t\right] + b_{ru}\right) \end{aligned}$$

$$\begin{aligned} rc^{t+1} &= \tanh\left(\Theta_{rc}\left[mr^{t+1}, \left(rr^{t+1} \odot rh^t\right)\right] + b_{rc}\right) \\ rh^{t+1} &= u^{t+1} \odot h^t + \left(1 - ru^{t+1}\right) \odot rc^{t+1} \end{aligned} \quad (4)$$

where h^t is the hidden state of the MGRU generated at the iteration step t and is also original hidden states of the RGRU for the iteration step $t + 1$, mr^{t+1} is the incoming message form other nodes, rh^{t+1} is the generated hidden state of the RGRU, rr^{t+1} and ru^{t+1} are the reset gate and update gate at time $t + 1$, respectively, and Θ_{rr} , Θ_{ru} , and Θ_{rc} are the learned parameters of filters in the RGRU.

Similar to RGRU, GGRU, and LGRU also take the incoming messages from object relationships (mr), global scene context (mg), and from local scene context (ml) as inputs and then output the hidden states hg and hl , respectively. In MGRU, the hidden states of nodes are affected by mr , mg , and ml jointly. The degree of influence on the hidden states of nodes may be varied. Inspired by the observations, we propose a novel weighted parametric-tensor-based fusion method that can fuse multiple hidden states of nodes in the MGRU. As discussed previously, MGRU has three different kinds of incoming messages (mr , mg , and ml). Each proposal node in the graph receives three kinds of messages and considers the messages jointly. The message encoding method and the weighted fusion method are discussed in the following.

1) *Messages Encoding From Spatial-Aware Object Relationships*: In this section, we discuss the messages encoding mechanism from spatial-aware object relationships. As discussed in the construction of the hierarchical scene graph, a directed edge $e_{i \rightarrow j}$, which is a scalar weight, denotes the influence flowing from node v_i to v_j . We utilize the visual feature fn_j and $e_{i \rightarrow j}$ together to denote the messages flowing from node v_i to v_j .

For each node, RGRU should receive messages encoding from other nodes. Therefore, we propose a pooling method to integrate messages from other nodes for each. Then, the RGRU will choose parts of the integrated message to update the hidden states of each node. Overall, the total incoming messages at the first iteration can be formulated as follows:

$$\begin{aligned} mr_1^1 &= MPool_i(e_{j \rightarrow i} fn_j, j \in V) \\ mr^1 &= \{mr_1, \dots, mr_i, \dots, mr_n\} \end{aligned} \quad (5)$$

where mr_i is the incoming messages of RGRU for node v_i , mr is the total incoming messages, V is the set of nodes in the constructed graph, $MPool_i$ is the message pool for node i , in our implementation, we utilize max pooling to extract the most important message, and $e_{i \rightarrow j}$ denotes the influence flowing from node v_i to v_j .

After the first iteration, the hidden states of nodes will be changed. When the hidden states of the nodes updated, the relationships among objects will also be updated. We utilize the updated hidden states of each node to compute the new messages from other nodes for the following iterations:

$$mr_i^t = MPool_i(e_{j \rightarrow i} h_j, j \in V), t > 1 \quad (6)$$

where t is the index of the iteration and h_j means the current hidden states of node v_j .

2) *Messages Encoding From Hierarchical Scene Context*: In this section, we discuss the messages encoding mechanism from the hierarchical scene context. As discussed in Section III-C, the entire image is set as the global scene context. For the messages from global scene context for node v_i , the entire image visual feature fsg is extracted as the global scene representation as the incoming message mg_i for v_i . In SKTCD, the messages from the global scene are not updated in all the iterations. Therefore, the incoming messages mg for the GGRU in all the iterations are as follows:

$$mg^t = \{mg_1^t, \dots, mg_n^t\} = \{fsg, \dots, fsg\} \quad (7)$$

where n is the number of nodes and fsg is the visual feature of the entire image.

For the messages from the local scene context of node v_i , we extract the visual feature fsl_i from the spatial information of the local scene context sl_i as the incoming messages in the first iteration. The incoming messages ml^1 of the first iteration for the LGRU are formulated as follows:

$$ml^1 = \{ml_1^1, \dots, ml_n^1\} = \{fsl_1, \dots, fsl_n\}. \quad (8)$$

As discussed in Section III-C, for sl_i of node v_i , we first find the smallest proposal node v_k , which includes node v_i . If the proposal node cannot be found, a region that is a little bigger than v_i is utilized. After the first iteration, the hidden states of nodes will be changed. In the following iterations, we utilized the updated hidden states of the nodes as the new messages of the local scene context. For example, if the local scene context of node v_i is set as node v_k , the current hidden states of h_k are set as the incoming messages from the local scene context for node v_i .

3) *Weighted Fusion*: As discussed previously, the hidden states of nodes can be affected by multiple factors, including hierarchical scene context and object relationships with spatial dependencies. The influence degrees of these different factors may be varied and have different effects on different train components. We adopt a weighted fusion method to handle this problem. The equation of the weighted fusion method for hidden state at time t can be formulated as follows:

$$h = W_{hg} \otimes hg + W_{hl} \otimes hl + W_{hr} \otimes hr \quad (9)$$

where h denotes the fused hidden state, $\otimes$ is the Hadamard product (i.e., elementwise multiplication for tensors), hg , hl ,

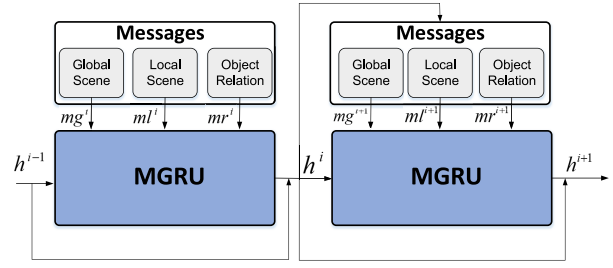


Fig. 7. Res-MGRU.

and hr are hidden states of GGRU, LGRU, and RGRU, respectively, and $W_{hg}, W_{hl}, \dots, W_{hr}$ are learnable parameters that can adjust the degrees affected by different factors.

Adopting the above formulation, the different hidden states of the GGRU, LGRU, and RGRU are fused together. The weights of fusing different hidden states can also be trainable. The fused hidden state is regarded as the new hidden state of MGRU.

F. Residual MGRU

A residual neural network is used with a reference to the direct hidden state, instead of an unreferenced function. The residual learning and linear shortcut connections can help to solve the exploding and vanishing gradient problems in the long-term backpropagation, especially in the networks with deep structures [46].

We introduce the residual connections into MGRU. An MGRU cell is considered as a computational block, where the residual information is passed by shortcuts of the blocks in order to speed up the convergence rate, as shown in Fig. 7. Hence, the hidden states of t with the residual shortcut path can be formulated as follows:

$$\tilde{h}_t = \phi(\tilde{h}_{t-1}, mg_t, ml_t, mr_t) + \tilde{h}_{t-1} \quad (10)$$

where ϕ denotes the function of MGRU cell, and $\tilde{h}_t$ denotes the state at time t with the residual shortcut path.

Equation (11) is composed of a shortcut connection and an elementwise addition. The previous hidden states of MGRU units are added to the next hidden states of the MGRU without nonlinear activation.

These residual connections can make the loss function close to an identity mapping as discussed in [46] and hence, the error is controlled within an acceptable range since the recurrent connections are formulated as identity mappings. Therefore, adopting the Res-MGRU can ensure maximum information flow between iterations, and hence make the initial hidden states (visual features of the nodes) have more influences on the eventually hidden states. This reconstruction can control the influences from the hierarchical scene context and other nodes within an acceptable range.

G. Optimization

After iterative reasoning for structural knowledge with Res-MGRU, the eventually integrated node features of each

proposal node are used to predict object categories and bounding box offsets. Our proposed Res-MGRU is plugged in Faster RCNN [35], [41], and we utilize the same optimization mechanism adopted in Faster RCNN. Eventually, there are two kinds of outputs. The first outputs are discrete probabilities of all the node, $p = (p_0, \dots, p_K)$, over $K + 1$ categories. As usual, p is computed by a softmax over the $K + 1$ outputs of a fully connected layer. The second outputs are bounding-box regression offsets t^k of each node, where k is the index of the object class. Each node is labeled with a ground-truth class u and a ground-truth bounding-box regression target v . A multitask loss L on each labeled node is adopted to jointly train for classification and bounding-box regression

$$L = L_{cls}(p, u) + \lambda L_{loc}(t, v) \quad (11)$$

where $L_{cls}(p, u) = -\log pu$ is log loss for true class u , and L_{loc} is defined over a tuple of true bounding box regression targets for class u . Interested readers can refer to [6], [35], and [41] for more details about the multitask loss.

IV. EXPERIMENTAL RESULTS

A. Dataset Description

Our dataset contains 1130 images captured from moving trains and a total of 18 590 critical components from 23 categories manually labeled by bounding boxes. We keep all the images in a high resolution of 1400×1024 to reserve the details of tiny components in trains. To evaluate the performance, we randomly split the dataset of train images into two parts: 1) training and 2) testing datasets. The training dataset contains 14 599 bounding boxes and the testing dataset contains 3991 bounding boxes. To demonstrate the efficiency of our proposed framework, we divide 23 categories of train components into three types based on their sizes: 1) large components (i.e., wheel_a, wheel_b, brake, spring, plate_a, and plate_b); 2) medium components (i.e., tube, buc_a, buc_b, joint_a, joint_b, plate_c, plate_d, fixator, and bearing); and 3) small components (i.e., nut_a, screw_a, nut_b, screw_b, wire, loop, and plug).

We also comprehensively evaluate our proposed model on the popular object detection dataset PASCAL VOC [47]. VOC involves 20 categories. The VOC 2007 dataset consists of about 5000 training images and 5000 testing images while the VOC 2012 dataset includes about 11 000 training images and 11 000 testing images.

B. Compared Baselines

Two variants of our proposed framework are evaluated in our experiment,¹ which are SKTCD based on Faster-RCNN [1] and SKTCDF based on FPNs [25]. Faster-RCNN is called FRCNN for simplicity. In these variants, our proposed Res-MGRU is plugged into the existing frameworks of FRCNN and FPN. FRCNN is a representative and well-known two-stage-based object detection framework. FPNs [25] are based on the FRCNN pipeline. It utilizes a top-down

architecture with lateral connections and shows significant improvement in several applications such as object detection especially on small object detection [25], which is based on the FRCNN pipeline. We compare SKTCD and SKTCDF with original FRCNN and FPN. Moreover, other three kinds of state-of-the-art object detection frameworks are also compared. The first kind of framework is one-stage-based object detection framework: 1) YOLOv3 [48] is the latest and efficient one-stage-based object detection system. YOLOv3 uses a few tricks to improve training to improve performance, including multiscale predictions, a better backbone classifier. The second kind of framework includes FRCNN with other reasoning methods: 2) SIN [13] exploits global scene context and object relationships based on FRCNN to boost detection performance. SIN only considers the global scene context and aggregates the influence of global scene context and object relationship equally. The hyperparameters of the backbone of SIN and RPN are the same as FRCNN. 3) RN [18] processes a set of objects simultaneously through interaction between objects based on the attention mechanism. 4) HKRM [20] incorporates the reasoning routed by two kinds of knowledge forms: an explicit knowledge module for structured constraints that are summarized with linguistic knowledge. The hyperparameters in training follow the released code of [20]. The third kind of framework includes FPN with other reasoning methods: 5) SINF. We build SIN based on FPN for comparison.

C. Implementation Details

We implement our proposed model on Pytorch. All the models are evaluated on 8 Tesla P100 cards on a single server. For Res-MGRU in our proposed SKTCD and SKTCF, the number of iterations is set as 2, and the hidden nodes of the GRU are set as 4096 and the layer of GRU is set as 1. For FRCNN, the hyperparameters in training mostly follow [41]. The total number of proposed regions generated by RPN after NMS is 128. For FPN, the total number of proposed regions after NMS is also set as 128.

The methods that are based on FRCNN and FPN, including our proposed SKTCD and SKTCDF, are with the same hyperparameters to ensure fairness. For fair comparison, ResNet-101 [46] pretrained on ImageNet [49] is utilized as the backbone network for all the methods. We train 24 epochs for all the methods and report the best performance of all the methods. (Further training after 12 epochs will not increase the performance of baseline.)

The performance metric is mean average precision (mAP) that is widely utilized to evaluate the object detection framework [1], [25]. We only report mAP scores using intersection-over-union (IoU) thresholds at 0.5 for the purpose of comparison with other existing methods.

D. Overall Performance

Table I presents the overall results while Table II presents the detailed results of detecting all the train components mAP^s , mAP^m , and mAP^l denote the mAP of detecting small, medium, and large train components, respectively. We can observe that

¹The dataset as well as the source code can be downloaded at <https://github.com/smartprobe/SKCD>.

TABLE I
DETECTION PERFORMANCE OVERVIEW

Methods	mAP	mAP^s	mAP^m	mAP^l
FRCNN	59.94	15.95	73.42	98.21
SIN	60.83	15.75	74.82	99.81
RN	60.66	15.60	74.67	99.71
HCRM	60.91	16.12	74.94	99.60
SKTCD	61.23	17.72	75.21	98.16
FPN	83.22	78.83	89.56	79.51
SINF	82.44	75.45	90.02	78.94
SKTCDF	87.18	80.32	90.95	90.57

FRCNN achieves better performance than YOLOv3. Both RN and SIN obtain better performance than FRCNN. However, FRCNN-based methods do not perform well on the detection of very small components (e.g., screw, nut, and bolt). SKTCD based on FRCNN achieves the best performance compared with other FRCNN-based baselines, including FRCNN, RN, SIN, and HCRM. It demonstrates the effectiveness of our proposed method, which explores the structural knowledge to boost the performance of train component detection.

The adoption of FPN significantly improves performance especially on small and medium train components. However, the performance on large train components of FPN is worse than that of FRCNN. The performance of SINF is a little worse than that of an FPN. One possible reason is that SINF does not consider the hierarchical scene context. Nevertheless, the local scene is important for detecting train components, especially for small and medium components. Our proposed SKTCDF which explores hierarchical scene context and spatial-aware object relationship obtains the best performance compared with other FPN-based baselines on detecting small, medium, and large components, respectively. Specifically, mAP is improved from 83.2% to 87.1%, mAP^l is significantly improved from 79.5% to 90.5%, mAP^m is improved from 89.5% to 90.9%, and mAP^s is improved from 89.5% to 90.9%. On the whole, SKTCDF obtains the best mAP compared with other baselines.

Overall, from these results, we can observe that applying our proposed model on both FRCNN and FPN frameworks can improve the detection performance. It demonstrates the importance of exploring hierarchical scene context and spatial-aware object relationship for train component detection.

E. Time Complexity of Different Approaches

In this section, we study the time complexity of our proposed SKTCD compared with other baselines. For simplicity, only one P100 GPU is utilized for all the approaches. We only compare our SKTCDF with the existing methods, which have achieved relatively good results. Table III shows the running time of FPN, SINF, and SKTCDF in the training for one epoch and the average prediction time for one image. We can observe that both the training time and testing time of SINF and our proposed SKTCDF are longer than that of FPN. The training time and testing time of SKTCDF are shorter than

SINF. It shows that our SKTCDF can achieve better prediction performance than SINF with shorter running time.

F. Results on Other Benchmark Dataset

In these experiments, we conduct some ablation studies to evaluate our proposed model on VOC 2007 and VOC 2012 to demonstrate the effectiveness of exploring the structural knowledge on general object detection. During the training phase, the training datasets of VOC 2007 and VOC 2012 are combined together as the training dataset. Then, we evaluate the model on the testing datasets on VOC 2007 and VOC 2012, respectively. The detection results of our proposed method and other compared baselines are shown in Table IV. It can be clearly observed that the SKTCD, which is based on FRCNN, outperforms other baselines on FRCNN, while the SKTDF, which is based on FPN, also outperforms the original FPN and SINF. Fig. 8 illustrates some qualitative comparisons of FPN and SKTCDF.

G. Effectiveness of Exploring Structural Knowledge

In these experiments, we evaluate the effectiveness of exploring structural knowledge, which includes hierarchical scene context and spatial-aware object relationships. Four variants based on SKTCDF are explored in the experiment, which are listed as follows: 1) SKTCDF-OR denotes only exploring the object relationships without spatial dependencies. For this variant, only the RGRU is utilized to fuse the messages from the object relationships. The spatial information is not considered in RGRU; 2) SKTCDF-ORS denotes exploring the spatial-aware object relationships to improve the performance by utilizing the RGRU; 3) SKTCDF-GSC means only utilizing the GGRU to explore the global scene context; and 4) SKTCDF-LSC stands for only utilizing the LGRU to explore the local scene context.

Table V shows the performance of the four variants and our proposed SKTCDF. In FPN, the detection results are only based on the object's visual appearance. SKTCDF, which has integrated hierarchical scene context and spatial-aware object relationships into visual features of objects, achieves the best mAP . It demonstrates the effectiveness of exploring structural knowledge with our proposed Res-MGRU. SKTCDF-ORS performs better than SKTCDF-OR, which shows that the detection performance can be improved by considering the relative positions among objects.

Figs. 9–12 illustrate some qualitative comparisons of FPN between SKTCDF-OR, SKTCDF-ORS, SKTCDF-GSC, and SKTCDF-LSC, respectively.

H. Effectiveness of Res-MGRU

In this section, we evaluate the effects of our proposed weighted fusion and residual mechanism in Res-MGRU. Two variants based on SKTCDF are explored in the experiments: 1) SKTCDF-RES denotes adopting the residual mechanism without the weighted fusion and 2) SKTCDF-WF denotes adopting the weighted fusion without the residual mechanism.

The experimental results are shown in Table VI. We can observe that adopting the residual mechanism can improve

TABLE II
DETECTION PERFORMANCE OF ALL THE TRAIN COMPONENTS

Types	Components	YOLOv3	FRCNN	SIN	RN	HKRM	SKTCD	FPN	SINF	SKTCDF
Large	wheel _a	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
	wheel _b	100.0	100.0	100.0	100.0	100.0	100.0	90.9	90.9	90.9
	brake	100.0	90.9	100.0	100.0	100.0	100.0	100.0	100.0	100.0
	spring	99.7	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
	plate _a	71.1	100.0	100.0	100.0	100.0	98.6	13.6	9.9	70.3
	plate _b	97.2	99.7	99.2	98.3	97.6	90.0	72.7	72.7	81.8
Medium	bearing	100.0	90.5	100.0	100.0	100.0	100.0	100.0	100.0	100.0
	buc _a	4.0	0.0	0.0	0.0	0.0	1.3	82.4	81.7	83.3
	buc _b	86.3	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
	joint _a	0.0	84.9	100.0	84.9	100.0	90.2	90.9	100.0	99.3
	joint _b	0.0	0.0	0.0	0.0	0.0	1.5	64.0	63.3	68.1
	plate _c	99.9	100.0	100.0	0.0	100.0	100.0	100.0	100.0	100.0
	plate _d	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
	fixator	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
Small	tube	19.8	85.8	73.4	72.1	74.5	84.6	68.3	74.8	67.7
	nut _a	68.9	17.7	14.4	13.7	16.9	24.6	89.6	89.8	89.9
	screw _a	31.1	9.1	2.3	2.5	3.1	4.6	86.4	85.2	85.8
	nut _b	0.0	2.8	9.1	8.7	9.1	4.6	79.6	80.3	80.9
	screw _b	0.0	0.0	0.0	0.0	0.0	9.1	66.4	40.4	67.3
	wire	0.9	0.0	0.0	0.0	0.0	0.0	19.9	18.8	21.7
	loop	0.0	0.0	0.0	0.0	0.0	0.0	90.1	90.4	99.3
	plug	98.5	98.7	99.9	99.9	99.9	99.3	98.6	98.8	98.7
	bolt	40.1	0.0	0.0	0.0	0.0	0.0	100.0	100.0	100.0

TABLE III
TIME COMPLEXITY OF DIFFERENT APPROACHES

Methods	Training time (s)	Testing time (s)
	One epoch	One image
FPN	629	0.23
SINF	969	0.93
SKTCDF	935	0.51

TABLE IV
OVERALL PERFORMANCE OF OBJECT DETECTION
ON VOC 2007 AND 2012 on *mAP*

Methods	VOC 2007	VOC 2012
YOLOv3	57.28	29.93
FRCNN	73.23	70.43
RN	75.82	72.19
SIN	76.05	73.12
HKRM	78.03	74.27
SKTCD	79.83	75.24

detection performance. That is, because adopting the Res-MGRU can ensure maximum information flow between iterations and hence make the initial hidden states (visual features of the nodes) have most influences on eventual hidden states. This reconstruction can control the influences from the hierarchical scene context and other nodes within an acceptable range. The performance of SKTCDF-WF is better than FPN, which shows that the weighted fusion method can help to improve the detection performance. After adopting these

TABLE V
EFFECTIVENESS OF EXPLORING THE STRUCTURAL KNOWLEDGE

Methods	<i>mAP</i>	<i>mAP^s</i>	<i>mAP^m</i>	<i>mAP^l</i>
FPN	83.20	78.84	89.52	79.54
SKTCDF-ORS	86.50	78.82	91.74	86.07
SKTCDF-GSC	84.56	78.20	90.76	83.75
SKTCDF-LSC	85.52	78.91	91.44	88.00
SKTCDF	87.18	80.32	90.95	90.57

TABLE VI
EFFECTIVENESS OF RESIDUAL AND WEIGHTED FUSION MECHANISM

Methods	<i>mAP</i>	<i>mAP^s</i>	<i>mAP^m</i>	<i>mAP^l</i>
FPN	83.20	78.84	89.52	79.54
SKTCDF-RES	85.90	78.00	90.90	88.64
SKTCDF-WF	85.39	78.31	89.17	89.16
SKTCDF	87.18	80.32	90.95	90.57

two mechanisms together, the performance can be further improved.

I. Influence of Iteration Steps

We evaluate the performance of different numbers of iteration steps. As shown in Table VII, SKTCDF with two iteration steps achieves the highest performance and gradually obtains worse afterward. A possible reason is that the messages from hierarchical scene context and spatial-aware object rela-

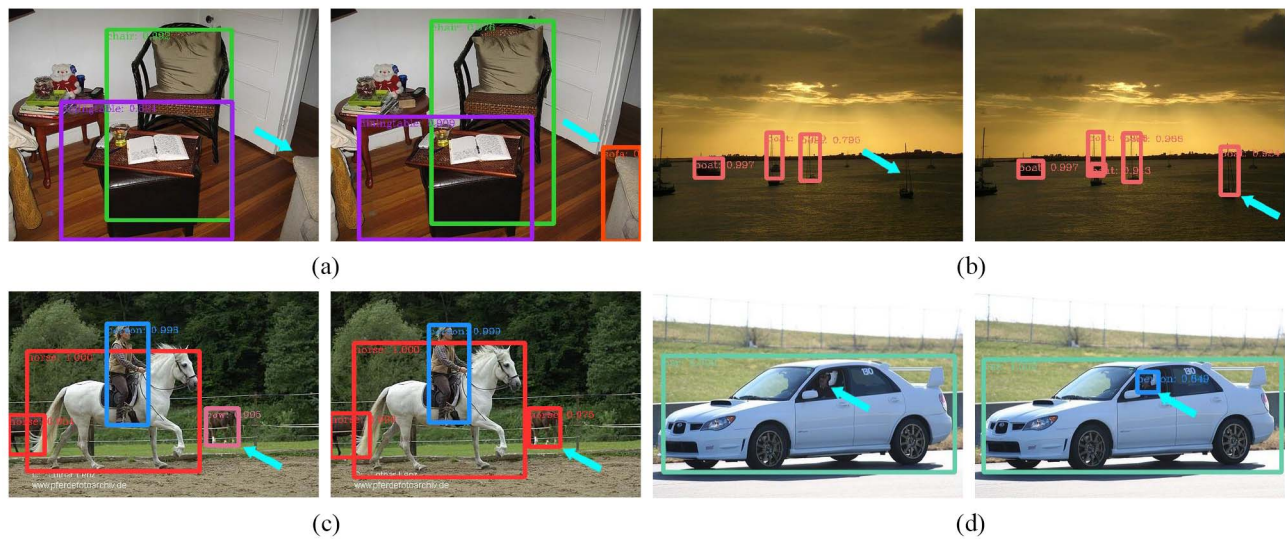


Fig. 8. Qualitative result comparison on FPN and SKTCDF. The left images in (a)–(d) are results on FPN, while the right images are results on SKTCDF. (a) It shows that a sofa is not detected by FPN, while SKTCDF can detect the sofa. (b) It shows that a boat is not detected by FPN, while SKTCDF can detect it. (c) It shows that a horse is misclassified as a cow, while SKTCDF can detect it correctly. (d) It shows that a person in a car is not detected, while SKTCDF can detect it.

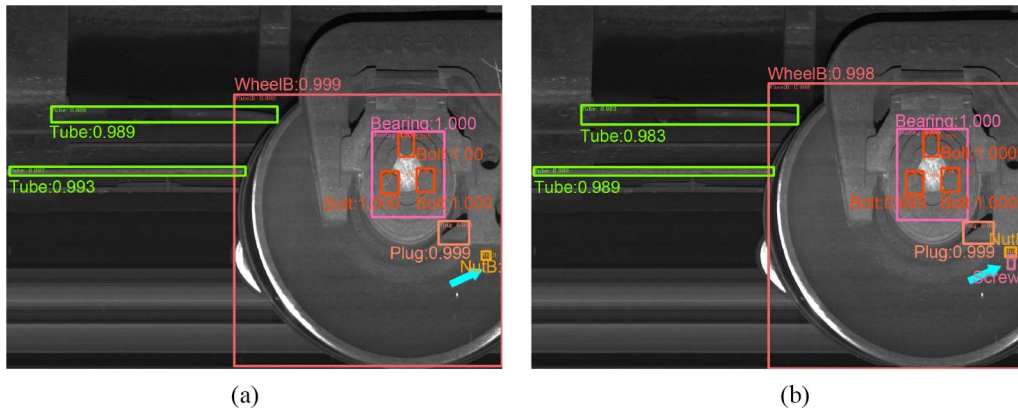


Fig. 9. Qualitative result comparison on FPN and SKTCDF-ORS. (a) It shows that FPN cannot detect a screw where the arrow indicates. (b) In contrast, SKTCDF-ORS can detect it by linking the relationship of nut and screw.

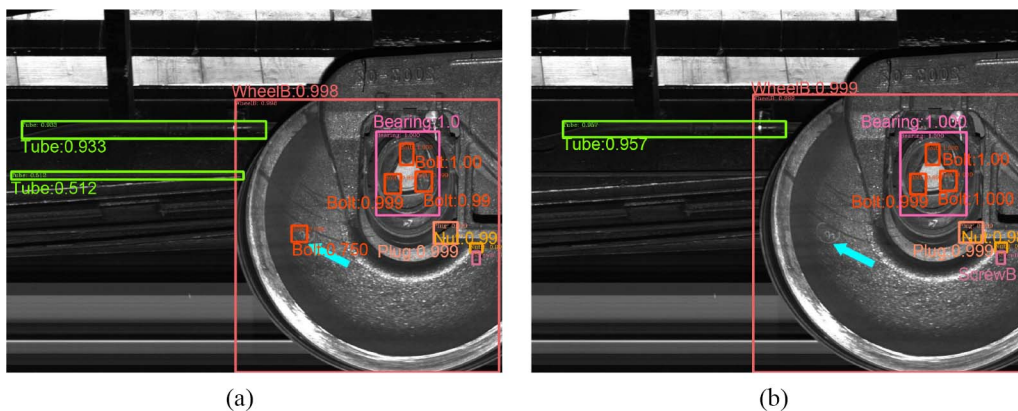


Fig. 10. Qualitative result comparison on FPN and SKTCDF-ORS. (a) Nut is mislabeled by FPN. (b) SKTCDF-ORS can avoid this kind of detection error through exploring the spatial-aware object relationships.

tionships are fused well within two iteration steps, after which noisy messages become significant and more disturbance is introduced.

V. CONCLUSION

In this article, we release a new train component dataset that contains high-resolution images of critical parts of moving

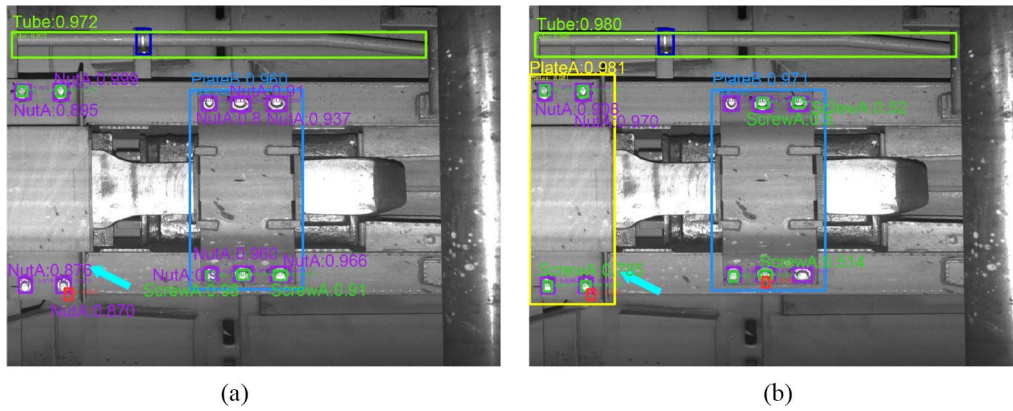


Fig. 11. Qualitative result comparison on FPN and SKTCDF-GSC. (a) FPN cannot detect a plate as shown in the left image. (b) In contrast, our proposed SKTCDF-GSC which consider the global scene context can detect the plate.

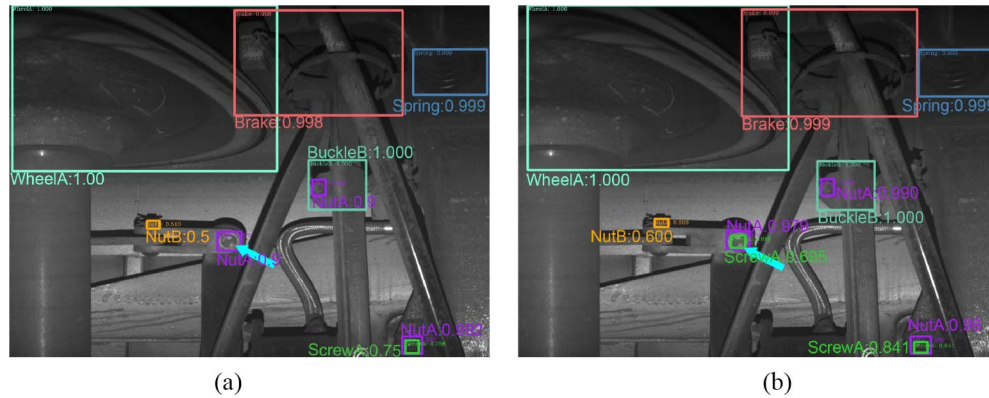


Fig. 12. Qualitative result comparison on FPN and SKTCDF-LSC. (a) FPN cannot detect a screw. (b) In contrast, our proposed SKTCDF-LSC which considers the local scene context for the screw can detect it.

TABLE VII
INFLUENCE OF THE ITERATION STEPS

Methods	AP	AP^s	AP^m	AP^l
Step-3	85.70	78.24	89.61	89.75
Step-4	85.97	79.52	90.39	87.93
Step-2	87.18	80.32	90.95	90.57

trains. The dominant object detection paradigm is limited by treating the visual features of each object region separately without considering common sense knowledge among objects. In this article, a novel end-to-end deep learning framework SKTCD has been proposed to take advantage of not only visual appearances but also hierarchical scene contexts and object relationships with spatial dependencies for train component detection. In SKTCD, we propose an Res-MGRU that can optimally fuse the visual features of train components and the encoded messages from the structural knowledge in a weighted-recurrent way. Extensive experiments on the newly constructed dataset and PASCAL VOC have demonstrated that SKTCD outperforms existing challenging baselines significantly.

In our future work, we aim to construct the explicit structural graph for the trains and then combine the constructed graph in deep learning framework to improve the performance of component detection for moving trains.

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